

3.10.4 Non-ionising imaging (A-level only)

3.10.4.1 Ultrasound imaging (A-level only)

Content

Reflection and transmission characteristics of sound waves at tissue boundaries, acoustic impedance, Z , and attenuation.

Advantages and disadvantages of ultrasound imaging in comparison with alternatives including safety issues and resolution.

Piezoelectric devices

Principles of generation and detection of ultrasound pulses.

A-scans and B-scans.

Examples of applications.

Use of the equations $Z = \rho c$ and $\frac{I_r}{I_i} = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2$

3.10.4.2 Fibre optics and endoscopy (A-level only)

Content

Properties of fibre optics and applications in medical physics; including total internal reflection at the core–cladding interface.

Physical principles of the optical system of a flexible endoscope; the use of coherent and non-coherent fibre bundles; examples of use for internal imaging and related advantages.

Basic principles of MR scanner;

- cross-section of patient scanned using magnetic fields
- protons initially aligned with spins parallel
- spinning hydrogen nuclei (protons) precess about the magnetic field lines of a superconducting magnet
- 'gradient' field coils used to scan cross-section
- short radio frequency (RF) pulses cause excitation and change of spin state in successive small regions
- protons excited during the scan emit RF signals as they de-excite
- RF signals detected and the resulting signals are processed by a computer to produce a visual image.

Students will not be asked about the production of magnetic fields used in an MR scanner, or about de-excitation relaxation times.